

Exploring the impacts of leader anger expressions on employee boundary-spanning behavior in construction projects

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Abstract

Purpose – Despite the increasing importance of boundary-spanning behaviors in construction projects, the research on how leader anger expressions impact employees' boundary-spanning behaviors remains unclear. This study aims to investigate the impact of leader anger expressions on employees' boundary-spanning behaviors in construction projects while exploring the mediating effect of work hope and the moderating effect of power distance orientation through the lens of social information processing theory.

Design/methodology/approach – The empirical data were collected from a questionnaire survey of 235 employees in construction projects, and the hypotheses were tested using the PROCESS program developed by Hayes.

Findings – This study demonstrated that leader anger expressions negatively affect employees' work hope, and work hope mediates the relationship between leader anger expressions and employees' boundary-spanning behaviors. Besides, the stronger the employees' power distance orientation, the stronger the negative relationship between leader anger expressions and employees' work hope, and the above mediating effect was also stronger.

Research limitations/implications – The use of cross-sectional data was unable to entirely infer a causal relationship between leader anger expressions and employees' boundary-spanning behaviors. Additionally, the reliance on self-reported data inevitably introduced some common method bias.

Originality/value – Applying social information processing theory, this paper enriches the discourse on factors influencing employees' boundary-spanning behaviors in construction projects. It also offers valuable insights for leaders in construction projects to manage their teams effectively.

Keywords Leader anger expressions, Work hope, Boundary-spanning behaviors, Power distance orientation

Paper type Research paper

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Ethical approval: All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

Informed consent: Informed consent was obtained from all individual participants included in the study.



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Introduction

As a cornerstone of global economic development, the construction industry typically operates on a project-based model (Liu *et al.*, 2021). Construction projects are temporary organizational endeavors characterized by clear time constraints, high uncertainty, and diverse talent (Bakker and Janowicz-Panjaitan, 2009; de Blois *et al.*, 2016; Liyanagamage *et al.*, 2022). With the integration of advanced technologies, compliance with stringent regulations, and the adoption of sustainable practices, construction projects have become increasingly complex and multifaceted (Regona *et al.*, 2022). These demands must be met within tight deadlines and budget constraints, posing significant challenges. To address these challenges effectively, employees in construction projects must engage in close interdisciplinary collaboration (Mell *et al.*, 2022b). In this context, it is vital for employees of construction projects to engage in boundary-spanning behaviors to dismantle knowledge barriers, enhance inter-member communication, and secure external support to address internal resource deficits and external challenges (Yan *et al.*, 2023). Employees' boundary-spanning behaviors predominantly involve interfacing with external entities to attain organizational goals (Zheng and Ahmed, 2024). Research has indicated that employees' boundary-spanning behaviors would foster knowledge integration (Mell *et al.*, 2022b), individual performance (Liu *et al.*, 2018), and creative behavior (Liu *et al.*, 2022). Hence, investigating the determinants influencing employees' boundary-spanning behaviors in construction projects is essential.

Existing research has examined how individuals' characteristics, organizational contexts, and leaders' behaviors influence employees' boundary-spanning behaviors in construction projects (Karrbom Gustavsson and Gohary, 2012; Cao *et al.*, 2021). Regarding individual characteristics, scholars have posited that individuals' nationality (Di Marco *et al.*, 2010), personal skills (Du and Pan, 2012), and richness of interpersonal networks (Brion *et al.*, 2012) are likely to affect individuals' boundary-spanning behaviors in construction projects. On organizational factors, research has suggested that organizational fraternization (Ramalingam and Mahalingam, 2011), organizational culture (Di Marco *et al.*, 2010) and organizational collaboration (Karrbom Gustavsson and Gohary, 2012) would influence members' boundary-spanning behaviors in construction projects. Given their roles as performance evaluators and resource custodians, leaders are pivotal in shaping employees' boundary-spanning behaviors (Marrone *et al.*, 2022). It is indicated that shared leadership (Cao *et al.*, 2021), vertical leadership (Bolden and Petrov, 2014), and leaders' supportive coaching behaviors (Marrone *et al.*, 2022) may also have an impact on employees' boundary-spanning behaviors in construction projects. However, current research on the impact of leader emotional expressions on employees' boundary-spanning behaviors is limited.

Compared to traditional organizations, project managers, as leaders, exhibit increased interaction frequency and enhanced relational proximity with employees (Rafique *et al.*, 2023). In construction projects with defined time constraints, employees lack the opportunity to deeply understand project managers' traits (Liyanagamage *et al.*, 2022). Their emotional expressions, due to the immediacy, directly influence employees' perceptions and behaviors (Silard and Dasborough, 2021). The high-stress nature of construction projects makes project managers' anger expressions common (Jiang *et al.*, 2023). Research in the realm of project management has found that leader anger expressions would reduce project members' proactive career behavior and leadership effectiveness while increasing their work alienation (Kang *et al.*, 2024; Lindebaum and Fielden, 2011). Yet, the effect of leader anger expressions on employees' boundary-spanning behaviors remains unclear. Therefore, this study aims to examine the impact of leader anger expressions on employees' boundary-spanning behaviors in construction projects, while further investigating the mediating mechanisms and boundary conditions of this effect.

This study utilizes social information processing theory to propose a moderated mediation model to explore the impact of leader anger expressions on employees' boundary-

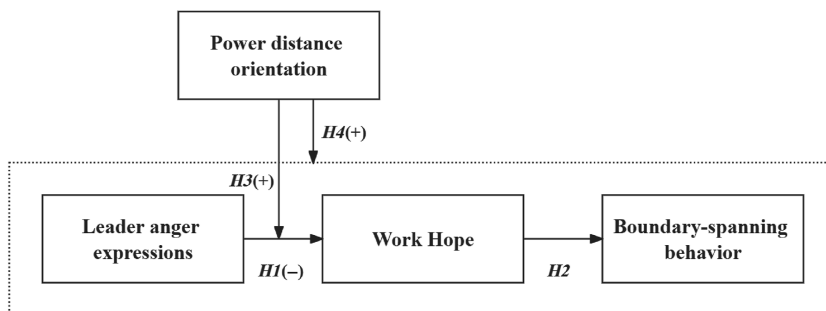
spanning behaviors. According to social information processing theory, leader anger expressions serve as pivotal social information, shaping employees' perceptions and behaviors (Wang *et al.*, 2022). Studies indicated that work hope, emerging from interpersonal dynamics, acts as an intermediate variable between leader emotional expressions and employees' behaviors (Wang *et al.*, 2023). Leader anger expressions may convey messages of incompetence and lack of value to employees, making them feel uncertain and unsupported (Shao and Martin, 2020), thereby reducing their work hope and boundary-spanning behaviors. Therefore, work hope is posited to mediate the influence of leader anger expressions on employees' boundary-spanning behaviors. Social information processing theory proposes that individual traits can affect their interpretation of social information (Lewicki, 1984). Power distance orientation signifies a key personal characteristic delineating an individual's level of acceptance of hierarchical disparity (Dorfman and Howell, 1988). Employees with high power distance orientation would likely perceive leader anger expressions as significant disapproval (Sundar and Bhaskar, 2021), impacting their boundary-spanning behaviors through work hope. Whereas those with low power distance orientation show less sensitivity to such cues (Peng *et al.*, 2021), resulting in less impact on their work hope and boundary-spanning behaviors.

This research aims to achieve three objectives. The first is to examine the impact of leader anger expressions on employees' boundary-spanning behaviors in construction projects. Through this objective, this study will expand the research on the factors impacting employees' boundary-spanning behaviors in construction projects. The second objective is to investigate the mediating role of work hope in the relationship between leader anger expressions and boundary-spanning behaviors. By exploring this objective, this study will enrich the mediating mechanisms through which leader anger expressions influence employees' boundary-spanning behaviors through the lens of social information processing theory. The final objective is to explore the moderating role of power distance orientation in the relationship between leader anger expressions and work hope. In fulfilling this objective, this study will broaden the boundary conditions in which leader anger expressions operate. Meanwhile, the results will also offer practical insights for organizations and leaders in construction projects on optimizing employee management strategies. The hypothetical model is presented in Figure 1.

Literature review

Social information processing theory

According to social information processing theory, individuals are surrounded by numerous cues in their living environment (Lewicki, 1984). Individuals interpret social cues from the



Source(s): Authors' own work

Figure 1.
The
hypothesized model

information base to generate cognition and behaviors (Salancik and Pfeffer, 1978). Employees rely more on social information, including leaders' emotions, to frame their evaluations in highly complex and unpredictable situations (Yang *et al.*, 2019). Construction work predominantly involves high-rise and open-air projects (Sundar and Bhaskar, 2021). The pervasive and destructive nature of leader anger expressions in construction projects necessitates research on its effects on employee cognition and behaviors (Liyanagamage *et al.*, 2022). In construction projects, leader anger expressions are more readily discernible and serve as social cues for employees. By internalizing these cues, employees develop perceptions that in turn guide their behaviors (Jiang *et al.*, 2023). Hence, this research applies social information processing theory to explore the influence of leader anger expressions on employees' cognition and behaviors.

Leader anger expressions in construction projects

Anger is the most readily discernible emotion, especially in construction projects where employees have limited time to deeply comprehend a leader's traits (Naderpajouh *et al.*, 2020). As direct responses to employees' behaviors and workplace events, leader anger expressions manifest not only through verbal articulations but also include non-verbal behaviors like expressive changes and physical gestures (Forgays *et al.*, 1997). Studies indicated that leader anger expressions are contingent upon the leader's personality traits and situational assessments (Shao and Guo, 2021). Stringent timelines, complex collaboration requirements, and pervasive safety imperatives of construction projects may precipitate heightened leader anger expressions (Irfan *et al.*, 2023).

Leader anger expressions constitute a critical research domain. Existing studies on leader anger expressions primarily focus on the realm of general management, while studies in construction projects only recently probed the mechanisms of their impacts on employees' cognition and behaviors. In the realm of general management, research has investigated the influence of leader anger expressions on employees' emotions (anger and anxiety, Eberly and Fong, 2013), cognition (perceived leader effectiveness, Yun *et al.*, 2020; perceived leader integrity, Shao and Martin, 2020), and behaviors (organizational citizenship behaviors, Koning and Van Kleef, 2015; innovative behaviors, Lee and Chang, 2014; voice behaviors, Song *et al.*, 2019). In the realm of construction projects, limited research has explored how leader anger expressions influence employees, and the discourse on leader anger expressions within project management remains underexplored. For example, Lindebaum and Fielden (2011) investigated the influence of leader anger expressions on employees' perceived leadership effectiveness in construction projects. Besides, Kang *et al.* (2024) explored the influence of leader anger expressions on newcomers' proactive career behavior and work alienation in construction projects. Considering the above, this study will further explore their impact on employees' boundary-spanning behaviors and the mechanisms of action.

Boundary-spanning behaviors in construction projects

Boundary-spanning behaviors denote employees' proactive efforts in transcending the demarcations of industry, organization, and knowledge, interfacing with the external environment to acquire resources and address challenges (Marrone *et al.*, 2007). Research revealed that employees' boundary-spanning behaviors are pivotal within construction projects for two reasons. Firstly, boundary-spanning behaviors will enhance interdisciplinary communication within construction projects characterized by diverse personnel (Wang *et al.*, 2022). Secondly, boundary-spanning behaviors will motivate project members to secure external support to navigate internal resource scarcity and external challenges (Cao *et al.*, 2021). Research has indicated that boundary-spanning behaviors in construction projects significantly enhance inter-project communication

(Stjerne *et al.*, 2019), team performance (Bartel-Radic and Munch, 2023), and trust building (Delozier and Burbach, 2021).

Investigating the determinants influencing boundary-spanning behaviors holds substantial theoretical and practical significance. Existing research in construction projects delves into the influence of project managers' higher education leadership (Pryor and Henley, 2018) and shared leadership (Cao *et al.*, 2021) on members' boundary-spanning behaviors. However, research on how leader emotional expressions, particularly anger expressions affect boundary-spanning remains scant. Building upon the above, this research scrutinizes how leader anger expressions influence employees' boundary-spanning behaviors through the lens of social information processing theory, with work hope serving as a mediator and power distance orientation serving as a moderator.

Research hypothesis

Leader anger expressions and employees' work hope

Work hope is the belief that individuals can find suitable ways to achieve their work goals, which determines their attitudes and behaviors (Snyder *et al.*, 1991). Given the high-pressure work environments in construction projects, cultivating work hope among employees helps them navigate stress effectively and fosters positive behaviors (Sundar and Bhaskar, 2021). Research has demonstrated that work hope is closely linked to interpersonal interactions (Stephanou and Athanasiadou, 2020). Due to intimate leader-employee contact, studies on the impact of leader attitudes and behaviors on employees' work hope are constantly expanding. Studies have verified how abusive, despotic, and destructive leadership behaviors affect employees' work hope (Avey *et al.*, 2022; Calheiros, 2018; Ahmad *et al.*, 2023).

Social information processing theory posits that leaders' emotions and behaviors are crucial information sources that employees process to develop their cognitions and behaviors (Salancik and Pfeffer, 1978). Leader anger expressions serve as pivotal social cues for employees, informing their interpretation of leaders' feedback and shaping subsequent perceptions (Shao and Guo, 2021). This research indicates that leader anger expressions impact work hope in two ways. Firstly, the social messages conveyed by leader anger expressions prompt employees to perceive a deficiency in achieving work goals (Liyangamage *et al.*, 2022), thereby reducing their work hope. Leader anger expressions in construction projects underscore the gravity and immediacy of issues, as minor errors during operations can precipitate significant negative repercussions (Peralta *et al.*, 2020). Once employees receive and internalize this social message, they may perceive that they are failing to attain their work goals (Stephanou and Athanasiadou, 2020), which may be detrimental to their work hope. Secondly, leader anger expressions also lead employees to perceive diminished worth and attenuate their work hope. Specifically, leader anger expressions convey messages to employees that they are perceived as lacking value to the organization (Avey *et al.*, 2022). Upon assimilating this message, employees will feel undervalued and unconfident (Emmerling *et al.*, 2023), thereby reducing work hope. In summary, the following hypothesis is proposed.

H1. Leader anger expressions negatively affect employees' work hope.

The mediating role of work hope

Boundary-spanning behaviors refer to activities undertaken by individuals beyond their formal roles or responsibilities (Marrone *et al.*, 2007). This research indicates that low work hope reduces employees' boundary-spanning behaviors. Firstly, employees with low work hope will curtail boundary-spanning behaviors by reducing the willingness to engage in searching, socializing, and task coordination. Individuals with low work hope often

experience confusion and uncertainty regarding their career prospects, exhibiting introverted dispositions (Stjerne *et al.*, 2019). As a result, they may lack the courage to explore unfamiliar territory (Delozier and Burbach, 2021), which leads to few boundary-spanning behaviors. Secondly, employees with low work hope may lack the conviction to perform competently and navigate unexpected events, thereby reducing boundary-spanning behaviors. They tend to feel worthless in the organization, resulting in a reduction of behaviors to demonstrate their team values (Stephanou and Athanasiadou, 2020). Boundary-spanning behaviors require employees to take proactive steps to validate their self-worth. Thus, based on self-consistency motivation (Bartel-Radic and Munch, 2023), low work hope among employees would lead to a reduction in boundary-spanning behaviors. Combining these two arguments, this study concludes that individuals with low work hope may reduce their boundary-spanning behaviors.

According to social information processing theory, leader anger expressions transmit information to employees. After interpreting the information, employees perceive themselves as lacking the capability to accomplish the work goal, which decreases their work hope. Low work hope reduces boundary-spanning behaviors. In summary, the following hypothesis is proposed.

- H2. Work hope mediates the relationship between leader anger expressions and employees' boundary-spanning behaviors.

The moderating effect of power distance orientation

Power distance orientation is a personality trait that reflects individuals' beliefs regarding the appropriateness of questioning authority (Dorfman and Howell, 1988). Individuals with varying power distance orientations perceive leaders' behaviors differently, resulting in distinct responses (Zhang *et al.*, 2023). Specifically, employees with high power distance orientation are distinguished by their adherence to hierarchical distinctions and readiness to align with leadership directives (Dorfman and Howell, 1988). While employees with low power distance prioritize equitable power distribution and advocate for parity between leaders and employees (Vitolla *et al.*, 2019).

This study proposes that employees' power distance orientation would strengthen the negative effect of leader anger expressions on employees' work hope. According to social information processing theory, power distance orientation, a critical personality trait, would lead to varied interpretations of leader anger expressions among employees, culminating in distinct perceptions and behaviors (Shao and Guo, 2021). Employees with high power distance orientation are highly attuned to hierarchical dynamics and exhibit heightened sensitivity to negative messages from leaders, such as anger expressions (Peng *et al.*, 2021). Such employees might construe leader anger expressions as heightened disapproval or discontent (Sundar and Bhaskar, 2021). This may exacerbate employees' feelings of skepticism towards their competency in meeting job expectations (Stephanou and Athanasiadou, 2020), consequently diminishing their work hope. Compared to high power distance orientation, employees with low power distance orientation exhibit less sensitivity to authority (Peng *et al.*, 2021). They have less concern for the social cues of leader anger expressions, indicating that leader anger expressions exert little negative influence on their work hope. In summary, the following hypothesis is proposed.

- H3. Employees' power distance orientation positively moderates the relationship between leader anger expressions and employees' work hope. The higher the employees' power distance orientation, the stronger the negative relationship between leader anger expressions and employees' work hope.

Based on the primary justification, we suppose that work hope mediates the relationship between leader anger expressions and employees' boundary-spanning behaviors. Furthermore, employees' power distance orientation positively moderates the relationship between leader anger expressions and work hope. When employees' power distance orientation is high, leader anger expressions show stronger negative effects on work hope. Hence, it is plausible to speculate that the research model proposed in this paper could demonstrate a moderated mediating effect. Accordingly, we propose.

- H4.* Employees' power distance orientation positively moderates the indirect effect of leader anger expressions on employees' boundary-spanning behaviors through work hope. The higher the employees' power distance orientation, the stronger the indirect effect.

Methods

A quantitative approach with the questionnaire survey was adopted in this study. The questionnaire's items were sourced from extensively validated academic publications. Then, we used convenience sampling to distribute questionnaires via the WeChat platform. After data collection, we utilized Mplus 7.0 to conduct confirmatory factor analysis for examining discriminant validity and SPSS 26.0 for descriptive analysis and regression analysis.

Measurement

The scales for this study were chosen based on the existing literature. Participants rated all items for leader anger expressions, work hope, boundary-spanning behaviors, and power distance orientation using a five-point Likert scale, with scores ranging from 1 (strongly disagree) to 5 (strongly agree).

Leader anger expressions. We applied a 5-item scale developed by [Forgays et al. \(1997\)](#) to measure leader anger expressions. One sample item is "At work, the project manager shows anger toward us". The Cronbach's α value was 0.91.

Work Hope. An 8-item scale developed by [Snyder et al. \(1991\)](#) was utilized to gauge work hope. One sample item is "I am capable of devising numerous strategies to resolve a difficult situation in the workplace". The Cronbach's α value was 0.91.

Boundary-spanning behaviors. Boundary-spanning behaviors were assessed using a six-item scale developed by [Marrone et al. \(2007\)](#). One sample item is "I will endeavor to convince external stakeholders, such as faculty and clients, to support project decisions". The Cronbach's α value was 0.86.

Power distance orientation. A six-item scale designed by [Dorfman and Howell \(1988\)](#) was used to assess power distance orientation. One sample item is "When making decisions, project managers do not need to seek my opinion". The Cronbach's α value was 0.73.

Control variables. Participants' age, gender, education, organizational tenure, tenure with the leader, and project type were included as control variables. These variables would affect participants' experiences, psychological perceptions, and coping behaviors ([Wang et al., 2018](#); [Mell et al., 2022a](#); [Lee et al., 2020](#)), thereby influencing the assessment of employees' boundary-spanning behaviors and perceptions of leader anger expressions. Therefore, we incorporated these variables as controls.

Sampling and data collection

This study employed convenience sampling, a method that allows researchers to gather data quickly and efficiently. The sample involved employees working on construction projects in China, selected from the author's extensive alumni network in the construction industry. These participants were chosen due to their accessibility and

high response rates (Leider *et al.*, 2024). The electronic link <http://www.wjx.cn> was used to create the electronic questionnaire, which was then distributed to participants via WeChat, a widely used communication app in China (Yang and Wang, 2021). The questionnaire assured respondents that their private information would not be released to third parties. And the data would be utilized only for research and aggregated for reporting purposes.

To ensure valid cases, two attention check items were included. The first item required respondents to select “agree”, and the second item required them to select “disagree”. The response options included 1 (strongly disagree), 2 (disagree), 3 (neutral), 4 (agree), and 5 (strongly agree). Through this approach, 20 were excluded because they did not select the “agree” or the “disagree” option of the specific items. Meanwhile, based on the principle of valid sample selection, 55 were excluded due to participants providing the same or regular answers. This study ultimately collected 235 valid cases. Based on the classification of construction projects by Liu *et al.* (2023), Wan *et al.* (2020), Yao *et al.* (2023) and Wang and Chen (2023), the project types in this study included civil engineering (47.2%), railway engineering (5.1%), municipal engineering (9.8%), mining engineering (6.4%), highway engineering (6.4%), civil aviation airport engineering (6.4%), telecommunications and radio and television engineering (7.7%), mechanical and electrical engineering (8.9%) and others (2.1%). Among the 235 valid respondents, 57.4% were female and were on average 25.22 years old (SD = 3.47). Their average tenure was 2.20 years (SD = 2.82) and the average tenure with the leader was 1.55 years (SD = 1.70). Also, 85.1% had received a bachelor’s degree and 14.9% had received a postgraduate degree or above.

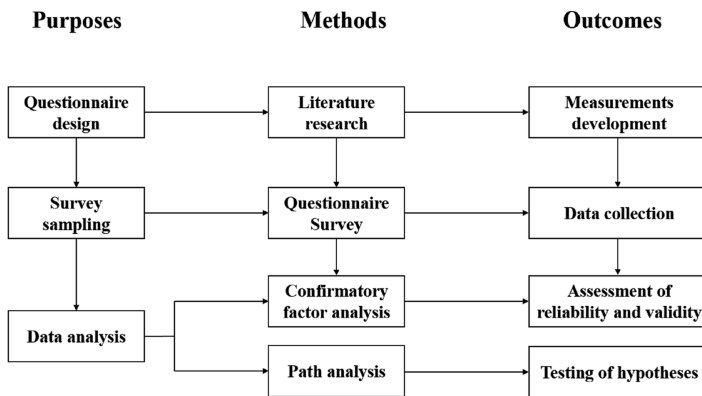
Analytical approach

After collecting the data, we performed the analysis with Mplus 7.0 and SPSS 26.0. Firstly, we utilized Mplus 7.0 to conduct a confirmatory factor analysis (CFA) to evaluate the discriminant validity of the measurements. Secondly, SPSS 26.0 was used to carry out descriptive analysis. Thirdly, we examined our hypotheses utilizing the PROCESS procedure of SPSS26.0 outlined by Hayes (2013). Hypothesis 1 and 2 were verified using PROCESS Model 4, and hypothesis 3 was verified using PROCESS Model 1. To mitigate potential multicollinearity concerns, we mean-centered leader anger expressions and employees’ work hope before testing hypothesis 3 (Aiken and West, 1991). Due to the non-normal distribution of indirect effects, scholars have recommended using bias-corrected (BC) bootstrap resampling to estimate the confidence intervals for the test of mediating effect (Hayes, 2009). Hypothesis 4 was verified using PROCESS Model 7. Quantification of moderated mediation effects involves a mathematical algorithm known as the “index of moderated mediation” (Hayes, 2015). Figure 2 illustrates the specific flow of research methods.

Results

We utilized Mplus 7.0 to perform confirmatory factor analyses (CFA) and assess the discriminant validity of the measurements. As depicted in Table 1, the hypothesized 5-factor model demonstrated a superior fit index compared to the alternative models, substantiating the discriminant validity of the measurements ($\chi^2 = 403.21$, $df = 269$, $RMSEA = 0.05$, $CFI = 0.95$, $TLI = 0.94$).

Table 2 shows a descriptive analysis of all variables and their bivariate correlations. The findings indicated statistically significant bivariate correlations between leader anger expressions and work hope ($r = -0.19$, $p < 0.01$), and work hope and boundary-spanning behaviors ($r = 0.49$, $p < 0.01$), providing initial support for hypotheses 1 and 2.



Source(s): Authors' own work

Figure 2.
Procedure of research
method

Model	χ^2	df	RMSEA	CFI	TLI
4-factor model	403.21	269	0.05	0.95	0.94
3-factor model	743.35	272	0.09	0.83	0.81
1-factor model	1671.57	275	0.15	0.48	0.44

Note(s): N = 235. 3-Factor model: Work hope and boundary-spanning behavior were collapsed onto one factor

Source(s): Generated by the authors based on primary research data

Table 1.
Results of
confirmatory factor
analyses

Variable	Mean	SD	1	2	3	4	5	6	7
1. Gender ^a	0.57	0.50							
2. Age	25.22	3.47	0.08						
3. Tenure	2.20	2.82	0.09	0.87**					
4. Tenure with leader	1.55	1.70	0.09	0.81**	0.90**				
5. Leader anger expressions	1.99	0.74	-0.12	0.07	0.01	0.10			
6. Work hope	3.61	0.70	-0.13	0.20**	0.18**	0.13*	-0.19**		
7. Power distance orientation	2.72	0.64	-0.17**	-0.01	0.03	0.03	-0.03	0.01	
8. Boundary-spanning behavior	3.65	0.62	-0.04	0.09	0.08	0.08	-0.36**	0.49**	0.09

Source(s): Generated by the authors based on primary research data

Table 2.
Means, standard
deviations, and
correlations of the
variables

Hypotheses tests

The findings of the hypothesis tests are displayed in [Table 3](#). Upon controlling for employees' demographic variables and project type, it was observed that leader anger expressions exhibited a negative correlation with work hope ($b = -0.21, p < 0.01$; Model 1, [Table 3](#)). Therefore, [hypothesis 1](#) was supported.

In [hypothesis 2](#), we validated that employees' work hope transmits the influence of leader anger expressions to employees' boundary-spanning behaviors. We assessed the mediating effect was examined using 5,000 bootstrapped samples and bias-corrected confidence intervals (CI). The statistics of [Table 3](#) revealed that the indirect effects of leaders (via work

Table 3.
Results of
hypothesis tests

Variable	Work hope		Boundary-spanning behavior	
	Model 1	Model 2	Model 3	Model 4
Gender	−0.24*	−0.29**	−0.12	−0.03
Age	0.05	0.07*	0.03	0.01
Tenure	0.01	0.01	−0.05	−0.05
Tenure with leader	−0.02	−0.06	0.07	0.08
Leader anger expressions (LAE)	−0.21**	−0.19**	−0.33**	−0.25**
Work hope (WH)				0.38**
Power distance orientation (PDO)		−0.06		
LAE × PDO		−0.40**		
<i>F</i>	5.66	5.62	8.44	18.33
<i>R</i> ²	0.33**	0.38**	0.39**	0.57**
Effect	Boot SE	LL 95% CI	UL 95% CI	
Bootstrap results for mediated effect				
−0.08	0.03	−0.14	−0.03	
Index of moderated mediation				
−0.15	0.05	−0.27	−0.06	
Note(s): <i>N</i> = 235. Unstandardized regression coefficients are reported. Bootstrap sample size = 5,000. LL = lower limit; CI = confidence interval; UL = upper limit				
* <i>p</i> < 0.05; ** <i>p</i> < 0.01				
Source(s): Generated by the authors based on primary research data				

hope) on boundary-spanning behaviors were statistically significant. Therefore, [hypothesis 2](#) can be supported (*effect* = −0.08, *boot SE* = 0.03, *95% CI* = [−0.14, −0.03]).

In [hypothesis 3](#), employees’ power distance orientation moderated the relationship between leader anger expressions and work hope, meaning that employees with high power distance orientation will have lower work hope than employees with low power distance orientation. The effect of leader anger expressions on work hope was significantly influenced by the interaction term with power distance orientation (*b* = −0.40, *p* < 0.01; Model 2, [Table 3](#)). We utilized the slope test proposed by [Aiken and West \(1991\)](#) to illustrate the interaction pattern more clearly, as shown in [Figure 3](#). Specifically, the moderating effect was not significant when the power distance orientation was one standard deviation below the mean (*b* = −0.64, *p* > 0.05), while it was significant when the power distance orientation was one standard deviation above the mean (*b* = 0.64, *p* < 0.01). Above indicated that the higher the employees’ power distance orientation, the stronger the negative relationship between leader anger expressions on employees’ work hope. Therefore, [hypothesis 3](#) was supported.

In [hypothesis 4](#), we proposed that leader anger expressions indirectly affect employees’ boundary-spanning behaviors, via work hope, which was stronger when employees’ power distance orientation was high. The last row of [Table 3](#) illustrated that moderated mediation was statistically significant (*effect* = −0.15, *boot SE* = 0.05, *95% CI* = [−0.27, −0.06]). Therefore, [hypothesis 4](#) received support.

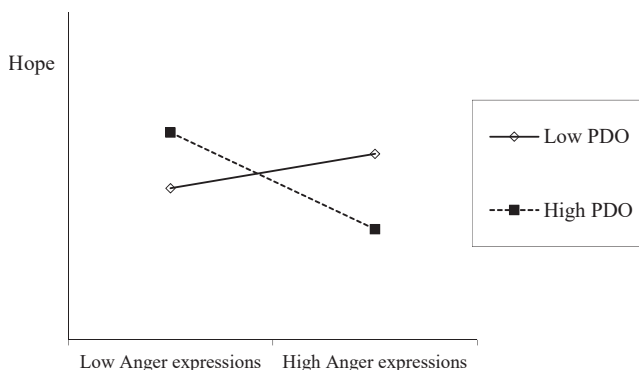
Discussion

This study utilized social information processing theory to explore how leader anger expressions in construction projects reduce employees’ boundary-spanning behaviors. As construction projects encounter increasingly complex environments ([Regona et al., 2022](#)), employees’ boundary-spanning behaviors have become essential, significantly enhancing

knowledge integration (Mell *et al.*, 2022b), individual performance (Liu *et al.*, 2018), and creative behavior (Liu *et al.*, 2022). Existing research has examined the impact of individual characteristics (Brion *et al.*, 2012), organizational contexts (Di Marco *et al.*, 2010), and leaders' behaviors (Cao *et al.*, 2021) on employees' boundary-spanning behaviors. Leaders, as decision-makers and performance evaluators, are pivotal influencers of these behaviors (Wang *et al.*, 2024). This study enriched previous research on leaders' influence on employees' boundary-spanning behaviors and further explored the effects of leader anger expressions on them.

According to social information processing theory, we confirmed that leader anger expressions negatively affect employees' work hope. Work hope mediated the relationship between leader anger expressions and employees' boundary-spanning behaviors. The results aligned with the previous research indicating that leaders' behaviors would significantly impact employees' boundary-spanning behaviors. For instance, shared leadership and vertical leadership in megaprojects (Cao *et al.*, 2021; Bolden and Petrov, 2014) and leaders' supportive coaching behaviors (Marrone *et al.*, 2022) in construction projects have been shown to facilitate employees' boundary-spanning behaviors. Similarly, this study demonstrated that leader anger expressions would also affect employees' boundary-spanning behaviors, but with a negative effect. The complexity of construction projects makes leader anger expressions quite common (Liyangamage *et al.*, 2022), while the intuitive nature of such expressions in construction projects means they are easily perceived as social information (Kang *et al.*, 2024), affecting employees' work hope and boundary-spanning behaviors. According to social information processing theory, employees faced with the message of leader anger expressions generate perceptions of low work hope, and they experience uncertainty about the future, which reduces their boundary-spanning behaviors.

In addition, we have demonstrated that employees' power distance orientation positively moderates the relationship between leader anger expressions and employees' work hope. Meanwhile, it also moderates the indirect effect of leader anger expressions on employees' boundary-spanning behaviors through work hope. The findings also echoed previous research. Suggesting that employees' power distance orientation would moderate the impact of leader negative emotional expressions and abusive leadership on employees (Chi and Ho, 2014; Peltokorpi and Ramaswami, 2021). Power distance orientation, encompassing individuals' views on hierarchical disparities, significantly shapes employees' discernment of social cues



Source(s): Created by the authors using original results

Figure 3.
The moderating role of
power distance
orientation on the
relationship between
leader anger
expressions and
work hope

pertaining to leaders' conduct (Dorfman and Howell, 1988). This study suggested that employees with high power distance orientation are more prone to interpret leader anger expressions as messages that devalue their self-worth (Shao and Guo, 2021), which reduces work hope. Consequently, the higher the employees' power distance orientation, the stronger the negative effect of leader anger expressions on employees' work hope.

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Theoretical implications

Firstly, this study expanded the findings regarding determinants of employees' boundary-spanning behaviors in construction projects. Previous studies have uncovered the influence of individual characteristics (Du and Pan, 2012; Brion *et al.*, 2012) and organizational contexts (Ramalingam and Mahalingam, 2011; Karrbom Gustavsson and Gohary, 2012) on project members' boundary-spanning behaviors. There are also studies that have explored the impact of leaders' behaviors (Cao *et al.*, 2021; Bolden and Petrov, 2014; Marrone *et al.*, 2022) on individuals' boundary-spanning behaviors in construction projects. This study extended the aforementioned findings by highlighting that leader anger expressions would significantly influence employees' boundary-spanning behaviors. Due to the pressing timelines and recurrent safety issues in temporary projects, leader anger expressions are commonplace and often yield negative behaviors for employees (Peralta *et al.*, 2020). Based on the above, we further demonstrated that leader anger expressions would influence employees' boundary-spanning behaviors, providing valuable theoretical insights for project management.

Secondly, this study delved into how leader anger expressions impact boundary-spanning behaviors through work hope from social information processing theory. Some scholars have used emotions as social information model (Shao, 2019a), self-determination theory (Schwarz Müller *et al.*, 2018), conservation of resources theory (Wang *et al.*, 2021), and leader-member exchange theory (Fisk and Friesen, 2012) to explain the effects of leader anger expressions on employees' behaviors. They respectively elucidated the mediating roles of members' anger intensity, follower-inferred negative intention, emotional exhaustion, and leader-member exchange quality. According to social information processing theory, we indicated that leader anger expressions signal to employees a deficiency in achieving objectives (Bartel-Radic and Munch, 2023), diminishing their work hope and in turn reducing their boundary-spanning behaviors.

Thirdly, we delineated the boundary conditions on the effect of leader anger expressions. Scholars have indicated that follower interdependence (Eberly and Fong, 2013), leader gender (Keck, 2019), vertical collectivism (Shao and Martin, 2020), and trust in the leader (Shao, 2019b) would moderate the impact of leader negative emotional expressions on employee cognition and behaviors. Construction projects facilitate close communication between leaders and employees, implying that employees' power distance orientation can significantly influence their perception of leader anger expressions (Sydow and Windeler, 2020; Yan *et al.*, 2020). This study indicated that employees with high power distance orientation are more susceptible to leader anger expressions, thereby amplifying the negative impact of leader anger expressions on employees' work hope.

Practical implications

The study revealed the negative impact of leader anger expressions on employees within construction projects. Organizations and leaders may implement the following measures to mitigate these adverse outcomes.

Firstly, organizations are advised to ensure leaders recognize the adverse impacts of their anger expressions and leaders should implement measures to mitigate the damage. In the construction industry, where projects are temporary and high-stress (Bakker and Janowicz-Panjaitan, 2009), leaders can establish "cooling-off periods" (Dorison *et al.*, 2020), meaning

they should step back when problems arise and calmly assess whether the situation warrants anger. After emotional stabilization, leaders should address issues with a solution-focused, assertive, and yet non-confrontational approach. Additionally, organizations should conduct anger management workshops tailored for high-pressure environments which include role-playing scenarios common in construction projects to help leaders practice coping strategies (Sadagheyani *et al.*, 2021).

Secondly, leaders should enhance employees' meaningfulness at work in order to elevate their work hope, particularly in project contexts involving leader anger expressions. In construction projects, implementing team-based learning and exchange programs is crucial for leaders. These not only help employees acquire a broader skill set, reducing monotony but also maintain high engagement levels (Mlekus and Maier, 2021), thereby enhancing their work hope. Additionally, leaders can introduce new devices such as real-time progress dashboards to provide live updates on project milestones, individual contributions, and overall project status. Such transparency allows employees to see the immediate impact of their work (Gara *et al.*, 2021), fostering a sense of achievement and enhancing work hope.

Finally, organizations are advised to screen individuals with high power distance orientation during recruitment and provide training for existing employees with such orientation. In the recruitment process, organizations should assess candidates' ability to cope with leaders' negative emotional expressions, selecting those who can effectively manage such situations. This can be achieved through psychological assessments and scenario simulations (Adamovic *et al.*, 2022). In training, organizations should provide specialized stress management programs, teaching high power distance employees techniques. These include deep breathing, temporarily stepping away from the situation, and seeking peer support to help them remain rational in the face of leader anger expressions (Toussaint *et al.*, 2021), thereby mitigating negative impacts.

Research limitation

There are several limitations that merit consideration in this study. The first is the utilization of a cross-sectional research design. As the data were gathered contemporaneously, it was not possible to fully infer the causal relationship between leader anger expressions and employees' boundary-spanning behaviors. Future research should prioritize developing longitudinal studies or experimental methods to examine the effects of leader anger expressions on boundary-spanning behaviors. Secondly, the model's variables were evaluated based on self-reported data from employees, inevitably introducing common method variance and confounding the findings. Future investigations are suggested to employ a multi-source data collection strategy to reduce the common method variance.

Thirdly, the data collection of leader anger expressions primarily relies on reports from employees, which may reflect their subjective perceptions rather than an accurate representation of the actual leader emotional expressions. Future research could employ a more objective data collection approach, encompassing multiple subordinates' perceptions of leader anger expressions to obtain a more objective measurement.

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